



New algorithm for cloud

Google announces TCP BBR, a new networking algo that can improve network throughput by about 4% globally. <http://dgit.in/GoogleTCP>



Food pictures to recipes

MIT Researchers have developed a neutral network that can identify food and find the recipe from an image. <http://dgit.in/MITrecipe>

If you were to now change this value of greeting it would automatically reflect the change in the app. For now you can try this with your browsers console, by setting different values for `app.greeting`.

TEMPLATES

You've already seen how templates can fill in values inside tags, but they can do a lot more. It can fill in attributes like a link's target location (href), or an image's source (src). Remember, Vue creates bindings, so when you update any property in data, Vue will also re-render the affected parts of your app with the new values.

The curly-brace syntax you've seen till now cannot be used for binding to attributes, Vue has a different syntax for that. Here are two examples of how to bind attributes on an image:

```

```

The v-bind here is what Vue calls a directive, and it sets up a binding between the src attribute and the url property. Since it's very common to bind attributes to properties, there's a simpler alternative syntax, simply omit v-bind and keep the colon as you can see with :alt.

Bindings needn't just be to properties, they can be JavaScript expressions like `{{ first_name + last_name }}`, or `:href="' /article/' + id"`.

Even calling methods inside binding is fair game as long as your method is also defined. For instance:

```
<h2>Author: {{
uppercase(author) }}</h2>
```

This would call the uppercase method, pass it the author and render the result in place. Just like properties are looked up inside data, methods are looked up inside methods.

For data transformations like above, Vue has a better syntax called filters: `{{ some_string | uppercase | reverse }}`

This will pass some string through the uppercase method and then pass the result through the reverse method. The above would be equivalent to

`reverse(uppercase(some_string))` but hopefully you'll agree the filter syntax is better. The methods used as filter need to be declared inside filters instead of methods.

Remember the GitHub User search app we wanted to build? Let's look at some code from that to learn about loops and conditionals in Vue templates:

```
<li v-for="user in users"
:key="user.id">
  
  <a :href="user.html_
url">{{ user.login }}</a>
  <span v-if="user.type ==
'User'">(U)</span>
  <span v-else>(O)</span>
</li>
```

The v-for directive here will output this list element for each entry in the users array. The current element will be available as user. Often you'll also need access to the iteration counter, you can get that using (user, index) in users. When looping it's best to provide a :key binding that's unique for each instance.

The two spans at the end show an if-else block at work in Vue. The v-if directive tells Vue to render the element only if the expression evaluates to true. If there's also an element with v-else like above, that will be rendered instead in its place.

While we haven't used CSS classes here, Vue provides some convenience features when dealing with them. The class attribute can be bound to a variable just like other attributes, however unlike the actual attribute your variable doesn't need to be a string.

If you provide an object like `:class="{ primary: isPrimary, disabled: !enabled }"` and Vue will then apply the primary class if isPrimary is true, and apply the disabled class if enabled is false. You can also provide an array of strings and objects which Vue will combine to make the final class string.

With such long templates it's inconvenient to fit them in a single line. The traditional ways of getting multi-line strings in JavaScript aren't very

*CODING MATTERS



*Apache Spark With Structured Streaming

Apache Spark 2.2 has been released with Structured Streaming no longer experimental. The accompanying release of PySpark is also available in pypi.

<http://dgit.in/ApacheSSS>



*Stack Overflow Channels In Beta

Stack Overflow's new product. Channels is targeted at teams of developers within companies who want a secure, private space to ask and answer proprietary questions.

<http://dgit.in/SOChanls>



*Python Heads IEEE Spectrum Language Ranking

The language that comes top in this year's IEEE Spectrum Ranking is Python, closely followed by C and Java. Use the interactive tool from IEEE to find out more.

<http://dgit.in/PythonWin>